

Derivation of the Korteweg-Helmholtz Electric and Magnetic Force Densities Including Electrostriction and Magnetostriction from the Quasistatic Poynting's Theorems

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Abstract: The Korteweg-Helmholtz electric and magnetic force densities for quasistatic systems including the Coulomb-Lorentz force density on free charge and free currents; polarization and magnetization force densities; and electrostriction and magnetostriction force densities are derived from Poynting's theorem. In this derivation, stationary frame linear constitutive laws are used for dielectric, magnetic, and ohmic conducting media with dielectric permittivity $\epsilon(\rho)$ and magnetic permeability $\mu(\rho)$, which can be functions of mass density ρ , itself a function of position and time, and σ is the ohmic conductivity which can be space and time varying. Using the quasistatic Galilean transformations for electric and magnetic field variables with conservation of mass and Poynting's theorem, results give a modified Poynting vector, differing from the usual $\vec{E} \times \vec{H}$ by a compressibility term, but with the usual electroquasistatic and magnetoquasistatic energy densities, power dissipation density, and Korteweg-Helmholtz force densities straight forwardly derived.

I. Background

There has been a confusion in the literature for the correct force density on compressible dielectric and magnetizable media [1]. The usually accepted force densities are the Korteweg-Helmholtz force density [1-5], valid for mass density (ρ) dependent constitutive relations, $\vec{D} = \epsilon(\rho)\vec{E}$ and $\vec{B} = \mu(\rho)\vec{H}$; and the Kelvin force density, based on non-interacting electric and magnetic dipole models, with arbitrary constitutive laws [1,2,6]. However, when the Kelvin force density is applied to media with linear constitutive laws, the Kelvin and Korteweg-Helmholtz force densities differ by the gradient of a scalar quantity. When applied to an incompressible medium, the total force on linear constitutive law media is the same for Kelvin and Korteweg-Helmholtz force densities, but the correctness of the force densities for compressible media seem to be not well understood.

Most derivations of the Korteweg-Helmholtz force density involve complicated perturbation analysis of energies, volume and surface integrations over all space, and complicated explanations [1-3]. However, this work will start from the differential form of the electroquasistatic (EQS) and magnetoquasistatic (MQS)

Minkowski forms of Maxwell's equations with linear constitutive laws corrected for motion using Galilean transformations. Then, using conservation of mass, a generalized Poynting's theorem is derived that gives a modified Poynting vector, differing from the usual $\vec{E} \times \vec{H}$ by a compressibility term, but with the usual EQS and MQS energy densities and power dissipation density, and most importantly the EQS and MQS Korteweg-Helmholtz force densities.

II. Minkowski Formulation of Maxwell's Equations

The relativistically correct Minkowski form of Maxwell's equations are:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad (\text{Faraday's Law}) \quad (1)$$

$$\nabla \times \vec{H} = \vec{J}_f + \frac{\partial \vec{D}}{\partial t} \quad (\text{Ampere's Law with Maxwell's Displacement Current Correction}) \quad (2)$$

$$\nabla \cdot \vec{D} = \rho_f \quad (\text{Gauss' Law for Electric Field}) \quad (3)$$

$$\nabla \cdot \vec{B} = 0 \quad (\text{Gauss' Law for Magnetic Field}) \quad (4)$$

The analysis is restricted to media with linear constitutive laws in a stationary reference frame

$$\vec{D} = \epsilon(\rho)\vec{E}, \quad \vec{B} = \mu(\rho)\vec{H}, \quad \vec{J}_f = \sigma\vec{E} \quad (5)$$

where $\vec{E}$ is the electric field, $\vec{B}$ is the magnetic flux density, $\vec{H}$ is the magnetic field intensity, $\vec{J}_f$ is the free volume current density, ρ_f is the free volume charge density, $\epsilon(\rho)$ is the dielectric permittivity and $\mu(\rho)$ is the magnetic permeability both dependent on the media mass density ρ , and σ is the ohmic conductivity where ρ and σ can be a function of position and time. The mass density ρ satisfies the conservation of mass condition

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0 \quad (6)$$

where $\vec{v}$ is the inertial frame velocity of the medium.

III. Quasistatic Formulations

Maxwell's Equations: At low frequencies, such that the electromagnetic wavelength, $\lambda = c/f$, is much larger than the size of the system, long time constants compared to the electromagnetic wave transit time

across the system, and for velocities v much less than the speed light c , $v \ll c$, Maxwell's equations divide into an approximate electroquasistatic (EQS) part where the electric fields $\bar{E}$ and $\bar{D}$ dominate while $\bar{H}$ and $\bar{B}$ are very small and an approximate magnetoquasistatic (MQS) part where the magnetic fields $\bar{H}$ and $\bar{B}$ dominate while $\bar{E}$ and $\bar{D}$ are small [6],

$$\begin{array}{ll} \text{EQS} & \text{MQS} \\ \nabla \times \bar{E} \approx 0 & \nabla \times \bar{E} = -\frac{\partial \bar{B}}{\partial t} \end{array} \quad (7) \quad (11)$$

$$\nabla \cdot \bar{D} = \rho_f \quad \nabla \times \bar{H} \approx \bar{J}_f \quad (8) \quad (12)$$

$$\nabla \times \bar{H} = \bar{J}_f + \frac{\partial \bar{D}}{\partial t} \quad \nabla \cdot \bar{B} = 0 \quad (9) \quad (13)$$

$$\nabla \cdot \bar{B} = 0 \quad \nabla \cdot \bar{D} = \rho_f \quad (10) \quad (14)$$

Galilean transformations: Because the quasistatic forms of Maxwell's equations are no longer relativistically correct, relationships between time and space now obey Galilean transformations rather than Lorentz transformations [7]. Then the current density constitutive laws for EQS and MQS ohmic media moving at inertial frame velocity $\bar{v}$ relative to the stationary coordinate system where (5) is valid are:

$$\begin{array}{ll} \text{EQS} & \text{MQS} \\ \bar{J} = \sigma \bar{E} + \rho_f \bar{v} & \bar{J} = \sigma (\bar{E} + \bar{v} \times \bar{B}) \end{array} \quad (15) \quad (16)$$

Note that for EQS systems that $\bar{D}$ and $\bar{E}$ do not change and $\bar{B}$ and $\bar{H}$ do change between Galilean inertial reference frames while for MQS inertial reference frames $\bar{H}$ and $\bar{B}$ do not change and $\bar{E}$ and $\bar{D}$ do change between Galilean inertial reference frames.

IV. EQS Poynting Theorem

Compressible Dielectric Media: We start with the vector identity

$$\nabla \cdot (\bar{E} \times \bar{H}) = \bar{H} \cdot (\nabla \times \bar{E}) - \bar{E} \cdot (\nabla \times \bar{H}) \quad (17)$$

Using the EQS Maxwell's equations in (7)-(10) yields

$$\nabla \cdot (\bar{E} \times \bar{H}) = -\bar{E} \cdot (\nabla \times \bar{H}) = -\bar{E} \cdot \bar{J}_f - \bar{E} \cdot \frac{\partial \bar{D}}{\partial t} \quad (18)$$

Using the constitutive laws of (5) and (15) results in

$$\nabla \cdot (\bar{E} \times \bar{H}) = -\sigma \bar{E} \cdot \bar{E} - \rho_f \bar{E} \cdot \bar{v} - \bar{E} \cdot \frac{\partial}{\partial t} (\varepsilon(\rho) \bar{E}) \quad (19)$$

We expand the last term as

$$-\bar{E} \cdot \frac{\partial}{\partial t} (\varepsilon(\rho) \bar{E}) = -\varepsilon(\rho) \bar{E} \cdot \frac{\partial \bar{E}}{\partial t} - \bar{E} \cdot \bar{E} \frac{\partial \varepsilon(\rho)}{\partial t} \quad (20)$$

$$= -\varepsilon(\rho) \frac{\partial}{\partial t} \left(\frac{1}{2} \bar{E} \cdot \bar{E} \right) - \frac{1}{2} \bar{E} \cdot \bar{E} \frac{\partial \varepsilon(\rho)}{\partial t} - \frac{1}{2} \bar{E} \cdot \bar{E} \frac{\partial \varepsilon(\rho)}{\partial t} \quad (21)$$

We broke up the last term on the right of (20) into two equal parts in (21) so that the first two terms on the right side of (21) can form a total differential

$$-\bar{E} \cdot \frac{\partial}{\partial t} (\varepsilon(\rho) \bar{E}) = -\frac{\partial}{\partial t} (\varepsilon(\rho) \frac{\bar{E} \cdot \bar{E}}{2}) - \frac{\bar{E} \cdot \bar{E}}{2} \frac{\partial \varepsilon(\rho)}{\partial t} \quad (22)$$

We can further reduce the last term on the right of (22) using

$$\frac{\bar{E} \cdot \bar{E}}{2} \frac{\partial \varepsilon(\rho)}{\partial t} = \frac{\bar{E} \cdot \bar{E}}{2} \frac{d\varepsilon(\rho)}{d\rho} \frac{\partial \rho}{\partial t} = -\frac{\bar{E} \cdot \bar{E}}{2} \frac{d\varepsilon(\rho)}{d\rho} \nabla \cdot (\rho \bar{v}) \quad (23)$$

where we used (6). Further expanding the divergence operator in (23) results in

$$\begin{aligned} \frac{1}{2} \bar{E} \cdot \bar{E} \frac{d\varepsilon(\rho)}{d\rho} \nabla \cdot (\rho \bar{v}) &= \frac{1}{2} \bar{E} \cdot \bar{E} \frac{d\varepsilon(\rho)}{d\rho} [\rho \nabla \cdot \bar{v} + \bar{v} \cdot \nabla \rho] \\ &= \bar{v} \cdot \left[\frac{\bar{E} \cdot \bar{E}}{2} \frac{d\varepsilon(\rho)}{d\rho} \nabla \rho - \nabla \left[\frac{\rho}{2} \frac{d\varepsilon(\rho)}{d\rho} \bar{E} \cdot \bar{E} \right] \right] + \nabla \cdot \left[\frac{\rho}{2} \frac{d\varepsilon(\rho)}{d\rho} \bar{E} \cdot \bar{E} \bar{v} \right] \\ &= \bar{v} \cdot \left[\frac{\bar{E} \cdot \bar{E}}{2} \nabla \varepsilon - \nabla \left(\frac{\rho}{2} \frac{d\varepsilon(\rho)}{d\rho} \bar{E} \cdot \bar{E} \right) \right] + \nabla \cdot \left(\frac{\rho}{2} \frac{d\varepsilon(\rho)}{d\rho} \bar{E} \cdot \bar{E} \bar{v} \right) \end{aligned} \quad (24)$$

where we recognize that

$$\frac{d\varepsilon(\rho)}{d\rho} \nabla \rho = \nabla \varepsilon(\rho) \quad (25)$$

Substituting (24) into (19)-(23) expresses (19) as

$$\begin{aligned} \nabla \cdot (\bar{E} \times \bar{H}) &= -\sigma \bar{E} \cdot \bar{E} - \frac{\partial}{\partial t} \left[\varepsilon(\rho) \frac{\bar{E} \cdot \bar{E}}{2} \right] + \nabla \cdot \left[\rho \frac{d\varepsilon(\rho)}{d\rho} \frac{\bar{E} \cdot \bar{E}}{2} \bar{v} \right] \\ &\quad - \bar{v} \cdot \left[\rho_f \bar{E} - \frac{1}{2} \bar{E} \cdot \bar{E} \nabla \varepsilon + \nabla \left(\frac{1}{2} \rho \frac{d\varepsilon(\rho)}{d\rho} \bar{E} \cdot \bar{E} \right) \right] \end{aligned} \quad (26)$$

Then (26) can be rewritten as

$$\nabla \cdot \bar{S}_E = -P_E - \frac{\partial w_E}{\partial t} - \bar{F}_E \cdot \bar{v} \quad (27)$$

where

$$\bar{S}_E = \bar{E} \times \bar{H} - \frac{1}{2} \rho \frac{d\varepsilon(\rho)}{d\rho} \bar{E} \cdot \bar{E} \bar{v}; \quad P_E = \sigma \bar{E} \cdot \bar{E} \quad (28)$$

$$w_E = \frac{1}{2} \varepsilon(\rho) \bar{E} \cdot \bar{E}; \quad \bar{F}_E = \rho_f \bar{E} - \frac{1}{2} \bar{E} \cdot \bar{E} \nabla \varepsilon(\rho) + \nabla \cdot \left(\frac{1}{2} \rho \frac{d\varepsilon(\rho)}{d\rho} \bar{E} \cdot \bar{E} \right)$$

$\bar{S}_E$ is the EQS Poynting vector representing electric and mechanical power going in or out of the electrodes, P_E is the power dissipation per unit volume, w_E is the stored electric energy per unit volume, and $\bar{F}_E$ is the Korteweg-Helmholtz force per unit volume. Note in (28) that $\bar{S}_E$ is modified from the usual $\bar{E} \times \bar{H}$ by a compressibility term.

Incompressible Dielectric Media: Note that for an incompressible medium where $\nabla \cdot \bar{v} = 0$ that any pressure-like term which is the gradient of a scalar quantity, ∇p , can be added to $\bar{F}_E$ with a corresponding change in $\bar{S}_E$ of $-p\bar{v}$. However, the total force on an incompressible medium will be unchanged. This is shown by integrating ∇p as the last term of (27) over all space including dielectrics and surrounding free space

$$\int_V \nabla p \cdot \bar{v} dV = \int_V \nabla \cdot (p\bar{v}) dV = \oint_S p\bar{v} \cdot \bar{da} \quad (29)$$

The surface S in (29) is bounded by rigid electrodes so that $\bar{v} \cdot \bar{da} = 0$. The remaining surface is at infinity and the last integral in (29) is zero if

$$\lim_{r \rightarrow \infty} \oint_S p\bar{v} \cdot \bar{da} = 0 \quad (30)$$

For example, the Kelvin force density for a linear dielectric is

$$\begin{aligned} \bar{F}_K &= \rho_f \bar{E} + (\bar{P} \cdot \nabla) \bar{E} = \rho_f \bar{E} + (\epsilon - \epsilon_0)(\bar{E} \cdot \nabla) \bar{E} \\ &= \rho_f \bar{E} + \frac{1}{2}(\epsilon - \epsilon_0) \nabla(\bar{E} \cdot \bar{E}) \\ &= \rho_f \bar{E} + \nabla \left(\frac{1}{2}(\epsilon - \epsilon_0) \bar{E} \cdot \bar{E} \right) - \frac{1}{2} \bar{E} \cdot \nabla \epsilon \end{aligned} \quad (31)$$

If we take

$$p = p_E = \frac{1}{2}[\epsilon(\rho) - \epsilon_0 - \rho \frac{d\epsilon(\rho)}{d\rho}] \bar{E} \cdot \bar{E} \quad (32)$$

then

$$\bar{F}_T = \bar{F}_E + \nabla p_E = \rho_f \bar{E} + (\epsilon - \epsilon_0)(\bar{E} \cdot \nabla) \bar{E} = \bar{F}_K \quad (33)$$

If $\bar{E}$ decreases with radial distance faster than $1/r$ as the surface extends to infinity, then (30) is satisfied

$$\lim_{r \rightarrow \infty} \oint_S \frac{1}{2}[\epsilon(\rho) - \epsilon_0 - \rho \frac{d\epsilon(\rho)}{d\rho}] \bar{E} \cdot \bar{E} \bar{v} \cdot \bar{da} = 0 \quad (34)$$

Note that even if the velocity $\bar{v}$ is finite at infinity, for an EQS system with zero net charge, $\bar{E}$ dies off as a dipole field as $1/r^3$ while the area increases as r^2 so that (34) goes to zero as $1/r^4$. The integral in (34) would also be zero even if one or more electrodes are at infinity so that the system for $r < \infty$ has net charge so that $\bar{E}$ dies off with radial distance as a point charge, $1/r^2$, so that (34) goes to zero as $1/r^2$.

This shows that using the Kelvin or Korteweg-Helmholtz force densities or any such force density supplemented with an arbitrary ∇p_E where $p_E \bar{v}$ goes to zero faster than $1/r^2$ at infinity will give the same total force on an incompressible material.

V. MQS Poynting Theorem

Compressible Magnetizable Media: We start again with (17) with the MQS Maxwell's equations in (11)-(14) to obtain

$$\nabla \cdot (\bar{E} \times \bar{H}) = \bar{H} \cdot (\nabla \times \bar{E}) - \bar{E} \cdot (\nabla \times \bar{H}) = -\bar{H} \cdot \frac{\partial \bar{B}}{\partial t} - \bar{E} \cdot \bar{J}_f \quad (35)$$

Using the stationary medium constitutive laws of (5) with (16) written as

$$\bar{E} = \bar{J}_f / \sigma - \bar{v} \times \bar{B} \quad (36)$$

lets us rewrite (35) as:

$$\begin{aligned} \nabla \cdot (\bar{E} \times \bar{H}) &= -\bar{H} \cdot \frac{\partial}{\partial t} [\mu(\rho) \bar{H}] - \frac{\bar{J}_f \cdot \bar{J}_f}{\sigma} + \bar{J}_f \cdot (\bar{v} \times \bar{B}) \\ &= -\bar{H} \cdot \frac{\partial}{\partial t} [\mu(\rho) \bar{H}] - \frac{\bar{J}_f \cdot \bar{J}_f}{\sigma} - (\bar{J}_f \times \bar{B}) \cdot \bar{v} \end{aligned} \quad (37)$$

where we used the scalar triple product identity

$$\bar{J}_f \cdot (\bar{v} \times \bar{B}) = -\bar{J}_f \cdot (\bar{B} \times \bar{v}) = -(\bar{J}_f \times \bar{B}) \cdot \bar{v} \quad (38)$$

Following the methodology of (20)-(23), we expand the first term on the right of (37) as

$$\begin{aligned} -\bar{H} \cdot \frac{\partial}{\partial t} [\mu(\rho) \bar{H}] &= -\mu(\rho) \bar{H} \cdot \frac{\partial \bar{H}}{\partial t} - \bar{H} \cdot \bar{H} \frac{\partial \mu(\rho)}{\partial t} \\ &= -\mu(\rho) \frac{\partial}{\partial t} \left(\frac{1}{2} \bar{H} \cdot \bar{H} \right) - \frac{1}{2} \bar{H} \cdot \bar{H} \frac{\partial \mu(\rho)}{\partial t} - \frac{1}{2} \bar{H} \cdot \bar{H} \frac{\partial \mu(\rho)}{\partial t} \\ &= -\frac{\partial}{\partial t} \left(\frac{1}{2} \mu(\rho) \bar{H} \cdot \bar{H} \right) - \frac{1}{2} \bar{H} \cdot \bar{H} \frac{d\mu(\rho)}{d\rho} \frac{\partial \rho}{\partial t} \\ &= -\frac{\partial}{\partial t} \left(\frac{1}{2} \mu(\rho) \bar{H} \cdot \bar{H} \right) + \frac{1}{2} \bar{H} \cdot \bar{H} \frac{d\mu(\rho)}{d\rho} \nabla \cdot (\rho \bar{v}) \end{aligned} \quad (39)$$

where in the last equality we use the conservation of mass relationship of (6). As in (24), the last term on the right in (39) can be further expanded to

$$\begin{aligned} \frac{1}{2} \bar{H} \cdot \bar{H} \frac{d\mu(\rho)}{d\rho} \nabla \cdot (\rho \bar{v}) &= \frac{1}{2} \bar{H} \cdot \bar{H} \frac{d\mu(\rho)}{d\rho} [\rho \nabla \cdot \bar{v} + \bar{v} \cdot \nabla \rho] \\ &= \bar{v} \cdot \left[\frac{1}{2} \bar{H} \cdot \bar{H} \nabla \mu - \frac{1}{2} \rho \frac{d\mu(\rho)}{d\rho} \bar{H} \cdot \bar{H} \right] + \nabla \cdot \left[\frac{1}{2} \rho \frac{d\mu(\rho)}{d\rho} \bar{H} \cdot \bar{H} \bar{v} \right] \end{aligned} \quad (40)$$

where in analogy to (25) we recognize that

$$\frac{d\mu(\rho)}{d\rho} \nabla \rho = \nabla \mu(\rho) \quad (41)$$

We now use (38)-(41) in (37) to obtain

$$\begin{aligned} \nabla \cdot (\bar{E} \times \bar{H}) &= -\frac{\partial}{\partial t} \left[\frac{1}{2} \mu(\rho) \bar{H} \cdot \bar{H} \right] - \frac{\bar{J}_f \cdot \bar{J}_f}{\sigma} \\ &\quad - \bar{v} \cdot \left[\bar{J}_f \times \bar{B} - \frac{\bar{H} \cdot \bar{H}}{2} \nabla \mu + \rho \frac{d\mu(\rho)}{d\rho} \frac{\bar{H} \cdot \bar{H}}{2} \right] + \nabla \cdot \left[\rho \frac{d\mu(\rho)}{d\rho} \frac{\bar{H} \cdot \bar{H}}{2} \bar{v} \right] \end{aligned} \quad (42)$$

Then in analogy to (26)-(28), (42) can be rewritten as

$$\nabla \cdot \bar{S}_M = -P_M - \frac{\partial w_M}{\partial t} - \bar{F}_M \cdot \bar{v} \quad (43)$$

where

$$\bar{S}_M = \bar{E} \times \bar{H} - \frac{\rho d\mu(\rho)}{d\rho} \frac{\bar{H} \cdot \bar{H} \bar{v}}{2}; P_M = \frac{\bar{J}_f \cdot \bar{J}_f}{\sigma}; w_M = \frac{\mu(\rho) \bar{H} \cdot \bar{H}}{2}$$

$$\bar{F}_M = \bar{J}_f \times \bar{B} - \frac{1}{2} \bar{H} \cdot \bar{H} \nabla \mu(\rho) + \nabla \left(\frac{1}{2} \rho \frac{d\mu(\rho)}{d\rho} \bar{H} \cdot \bar{H} \right) \quad (44)$$

$\bar{S}_M$ is the MQS Poynting vector representing magnetic and mechanical power going in or out of the electrodes, P_M is the power dissipation per unit volume, w_M is the stored magnetic energy per unit volume, and $\bar{F}_M$ is the Korteweg-Helmholtz force per unit volume. Note again in (44) that $\bar{S}_M$ is modified from the usual $\bar{E} \times \bar{H}$ by a compressibility term.

Incompressible Magnetizable Media: Using (29)-(30) and analogous relations to (31)-(34), the Kelvin force density for an incompressible linear magnetic material can be obtained by adding a gradient of pressure term to $\bar{F}_M$ where

$$p = p_M = \frac{1}{2} [\mu(\rho) - \mu_0 - \rho \frac{d\mu(\rho)}{d\rho}] \bar{H} \cdot \bar{H} \quad (45)$$

Then

$$\begin{aligned} \bar{F}_T &= \bar{F}_M + \nabla p_M = \bar{J}_f \times \mu(\rho) \bar{H} - \frac{\bar{H} \cdot \bar{H}}{2} \nabla \mu(\rho) \\ &+ \nabla \left(\frac{1}{2} \rho \frac{d\mu(\rho)}{d\rho} \bar{H} \cdot \bar{H} \right) + \nabla \left(-\frac{1}{2} [\mu(\rho) - \mu_0 - \rho \frac{d\mu(\rho)}{d\rho}] \bar{H} \cdot \bar{H} \right) \\ &= \bar{J}_f \times \mu(\rho) \bar{H} - \frac{1}{2} (\mu(\rho) - \mu_0) \nabla (\bar{H} \cdot \bar{H}) \end{aligned} \quad (46)$$

Using the identity

$$\begin{aligned} \frac{1}{2} \nabla (\bar{H} \cdot \bar{H}) &= \bar{H} \cdot \nabla \bar{H} - (\nabla \times \bar{H}) \times \bar{H} \\ &= \bar{H} \cdot \nabla \bar{H} - \bar{J}_f \times \bar{H} \end{aligned} \quad (47)$$

reduces (46) to

$$\begin{aligned} \bar{F}_T &= \bar{J}_f \times \mu(\rho) \bar{H} + (\mu(\rho) - \mu_0) (\bar{H} \cdot \nabla \bar{H} - \bar{J}_f \times \bar{H}) \\ &= \mu_0 \bar{J}_f \times \bar{H} + (\mu(\rho) - \mu_0) \bar{H} \cdot \nabla \bar{H} \\ &= \mu_0 \bar{J}_f \times \bar{H} + \mu_0 \bar{M} \cdot \nabla \bar{H} = \bar{F}_K \end{aligned} \quad (48)$$

where $\bar{F}_K$ is the magnetic Kelvin force density. Note that the Lorentz force contribution to the Kelvin force density in (48) is $\mu_0 \bar{J}_f \times \bar{H}$, while the Lorentz force contribution to the Korteweg-Helmholtz force density in (44) is $\mu(\rho) \bar{J}_f \times \bar{H}$.

VI. Conclusion

By a straight forward use of the Minkowski formulation of Maxwell's equations in electroquasistatic and magnetoquasistatic limits to derive Poynting's theorem using only conservation of mass and Galilean transformations for the current density-electric field constitutive law, we were able to directly derive the Korteweg-Helmholtz force densities for compressible linear dielectric and magnetic media whose permittivity and magnetic permeability depend only on mass density, including Coulomb and Lorentz forces if the dielectric media is charged or if the magnetic media is carrying current. Adding a pure gradient of a scalar quantity to the Korteweg-Helmholtz force density for an incompressible material will not change the total force on a material. In addition, the Poynting vector power flow is modified from the usual $\bar{E} \times \bar{H}$ by a compressibility term.

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